Necessary size constraints for entirely ideal right-angled polytopes in dimensions five and six

Coxeter Polytopes Frontier Project

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Abstract

An entirely ideal finite-volume right-angled hyperbolic five-polytope has at least 63 facets and 211 vertices. In dimension six the corresponding bounds are 275 facets and 6,106 vertices, with $\mathrm{V}(P) \geq \lceil 6039\,\mathrm{F}(P)/272 \rceil$. The proof combines an actual outside-cusp neighbor capacity, fixed-scale polygon identities, an exclusion of an eleven-vertex equality carrier, and good-edge cusp transport with a sharp reciprocal estimate. One previously proved internal small-facet theorem is stated separately with its published three-dimensional classification input. With the published Nikulin–Khovanskii and Dufour inequalities, the entirely ideal five- and six-face fractions in ambient dimensions eleven and twelve are respectively less than 4/47 and 15/248. These are necessary size and incidence constraints. They neither exclude dimensions five or six nor supply examples or a full formal proof.

1 Scope, inputs and actual faces

All polytopes are full-dimensional, convex and finite-volume. An *entirely ideal* polytope has only ideal vertices. Write $\mathrm{V}(X)$ and $\mathrm{F}(X)$ for its vertex and facet counts. Faces of dimension at least three inherit finite volume and right angles; two-faces are ideal polygons. At an ideal vertex of an n-dimensional right-angled polytope the horospherical link is a rectangular $(n-1)$-box; its 2^{n-1} vertices give the graph degree. The polytope is simple at the relative interior of every positive-dimensional face. These standard geometric facts, and acuteness of all actual outward wall normals, are our foundational inputs.

We use the Lorentz model with future timelike vectors of negative square and outward unit spacelike normals n_i. Thus $P = \{x \in \mathbb{H}^n : \langle x, n_i \rangle \leq 0\}$ and distinct actual normals have nonpositive products, including facets not sharing a visible ridge. The following all-wall argument will be used throughout.

Lemma 1.1 (Acute projection and the wall-product gap)**.** *Suppose selected actual normals have positive-definite Gram matrix M with entrywise nonnegative inverse. Their common hyperplanes meet P in an actual face, with all other walls strict at a finite relative- interior point. In a right-angled polytope, distinct actual unit normals therefore have product either zero or at most -1. In an entirely ideal n-polytope, n mutually orthogonal actual normals cannot occur.*

Proof. Choose a strict future timelike interior point q. With $b_i = \langle q, n_i \rangle < 0$ and $a = M^{-1}b$, put $p = q - \sum_i a_i n_i$. Every $a_i < 0$, the selected products vanish, and

$$p^2 = q^2 - b^T M^{-1} b < 0, \qquad \langle p, n_j \rangle = \langle q, n_j \rangle - \sum_i a_i \langle n_i, n_j \rangle < 0$$

for any other actual wall. Also $\langle q, p\rangle < 0$, so p is future. After normalization it is the asserted actual point. In particular a two-root product in $(-1, 0)$ would give an actual nonright ridge, and n orthogonal roots would give an actual finite vertex. □

A positive null combination of actual outward normals is nonzero because its product with q is negative. Its ray is future and feasible against every other wall by acuteness. Compact sections or assertions about feasible edges below will always include this all-wall condition.

Lemma 1.2 (Facet injection)**.** *For an entirely ideal facet A of an entirely ideal right-angled P,*

$$\mathrm{F}(P) \geq 1 + \mathrm{F}(A) + \mathrm{V}(A). \tag{1}$$

Proof. Count A, its distinct ridge-neighbor facets, and at each vertex u of A the unique actual facet opposite A in the rectangular cusp link. A cusp-opposite hyperplane is asymptotic to A at u only, so it cannot serve two different cusps. Nor can it be a ridge neighbor, whose hyperplane is orthogonal to A. □

For an ideal right-angled three-polyhedron with m vertices, degree four and Euler give $2m$ edges and $m + 2$ polygonal faces. If p_k counts its k-gonal faces, then

$$p_3 - \sum_{k\geq 5}(k-4)p_k = 8. \tag{2}$$

In particular it has at least six vertices and eight triangular faces. These elementary identities need no classification.

2 The outside-cusp capacity

Lemma 2.1. *An ideal vertex outside a facet A of an entirely ideal right-angled n-polytope has at most two graph-neighbors in A. Consequently*

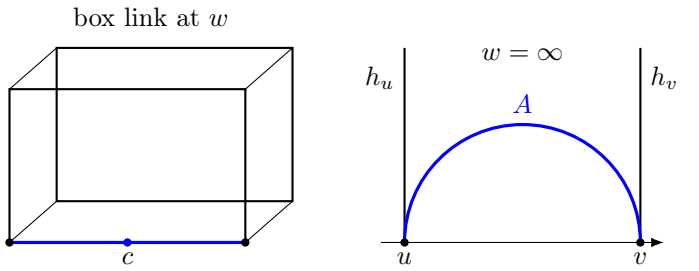
$$\mathrm{V}(P) \geq \left(1 + 2^{n-3}\right) \mathrm{V}(A). \tag{3}$$

Proof. Send the outside vertex w to infinity. Its incident vertical walls bound $\prod_{i=1}^{n-1}[0, L_i]$. The wall of A is a hemisphere of center c and radius R, with P outside its ball. Its unit-normal products with the two coordinate walls are $-c_i/R$ and $(c_i - L_i)/R$. Lemma 1.1 implies

$$c_i \in \{0\} \cup [R, \infty), \qquad L_i - c_i \in \{0\} \cup [R, \infty). \tag{4}$$

An edge from w is vertical over a box corner ξ. If its other endpoint lies on A, then $\sum_i (\xi_i - c_i)^2 = R^2$. Every nonzero summand's absolute coordinate difference is at least R by (4). Exactly one is nonzero, and equals R. Thus c lies on all but at most one boundary coordinate hyperplane. It cannot be a box corner: A and those $n - 1$ vertical walls would be mutually orthogonal actual walls, forcing an actual finite vertex. Therefore c is in the relative interior of one box edge; only its two endpoints can satisfy the sphere equation. If there is no neighbor, no assertion about the position of c is needed.

At a vertex of A, precisely 2^{n-2} edges leave A, by comparing its two box links. Their other endpoints lie outside A: an edge whose endpoints lie in a face lies in that face. Counting outgoing edges against the proved capacity gives $2^{n-2}\,\mathrm{V}(A) \leq 2(\mathrm{V}(P) - \mathrm{V}(A))$. □



only two contact corners on one box edge

tangency to opposite walls matches the other coordinates

Figure 1: The capacity argument (left, a box-link schematic) and the two-dimensional section used in the actual tangency argument (right). The latter is applied only after both tangency points are actual vertices.

3 Fixed-scale polygon identities

Let A be a facet of an ideal right-angled polytope, with unit normal n. At each vertex u let h_u be its actual cusp-opposite facet and put $t_u = n + h_u$. This is a future null vector perpendicular to n, with scale fixed by the two unit normals. Distinct u give distinct h_u. For an edge uv of A, its other supporting normals are orthogonal to both h_u, h_v. If these were orthogonal as well, Lemma 1.1 would give a finite ambient vertex. Hence

$$-\langle t_u, t_v\rangle = 1 - \langle h_u, h_v\rangle \geq 2. \tag{5}$$

At a cusp of A, each opposite pair of intrinsic unit facet normals has positive sum αt_u. A corner of a polygonal two-face uses one such pair; its intrinsic normals remain unit because they are orthogonal to the supporting walls. The following statements use this same scale at every corner, not freely rescaled null representatives.

Lemma 3.1 (Small corners)**.** *Every polygonal two-face of A has at least two corner coefficients $\gamma < 2$. If a triangular face has consecutive null products -2, all three coefficients are one.*

Proof. In the polygon's Lorentz plane use

$$T(a,b) = (a^2+b^2, a^2-b^2, 2ab), \qquad \langle T(v), T(w)\rangle = -2\det(v,w)^2.$$

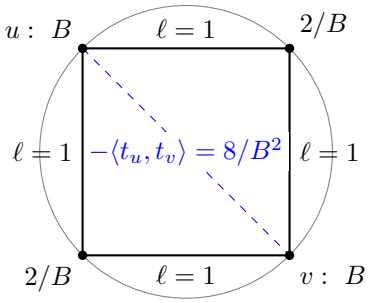
Choose cyclic projective lifts v_i with $t_i = T(v_i)$ and signs $v_0 = -v_k$, $v_{k+1} = -v_1$. The successive determinants $\ell_i = \det(v_i, v_{i+1})$, including the lifted last gap, are positive and at least one by (5). A direct side-normal calculation gives

$$\gamma_i = \frac{\det(v_{i-1}, v_{i+1})}{\ell_{i-1}\ell_i}, \qquad \frac{v_{i-1}}{\ell_{i-1}} + \frac{v_{i+1}}{\ell_i} = \gamma_i v_i. \tag{6}$$

For example send v_i to $(1,0)$ and write its neighbors as $(a, -\ell_{i-1})$, (b, ℓ_i). The outward side normals are $(a/\ell_{i-1}, a/\ell_{i-1}, -1)$ and $(b/\ell_i, b/\ell_i, 1)$; their sum is exactly $\gamma_i T(1,0)$.

The symmetric recurrence matrix has diagonal γ_i, off-diagonal $-1/\ell_i$, and positive wrap entries $1/\ell_k$. Its kernel contains the two independent spinor coordinate arrays. If at most one diagonal were below two, delete that index, or any index if none is below two. The remaining path matrix is positive definite: its quadratic form dominates

$$x_1^2 + x_{k-1}^2 + \sum_{j=1}^{k-2} (|x_j| - |x_{j+1}|)^2 > 0$$



$\langle h_u, h_v\rangle = 0$ forces $B = 2\sqrt{2}$;
the two other coefficients are $1/\sqrt{2}$.

Figure 2: A decorated ideal quadrilateral, drawn in a schematic Klein disc. Coefficients use the actual cusp-opposite normalization. Its opposite-product identity will exclude the equality carrier.

for nonzero x. Its rank $k-1$ contradicts the two kernel vectors. This includes $k = 3$, with the indicated wrap sign. For a triangle with all $\ell_i = 1$, formula (6), or the three pair-sum products, gives $\gamma_i\gamma_j = 1$ and hence all $\gamma_i = 1$. □

Lemma 3.2 (A decorated quadrilateral)**.** *If all four consecutive null products on a quadrilateral equal -2, its corner coefficients alternate $B, 2/B, B, 2/B$. For the two opposite corners with coefficient B,*

$$-\langle t_u, t_v\rangle = 8/B^2. \tag{7}$$

Proof. After a determinant-one spinor change the cyclic lifts are $(1,0),(0,1),(-1,t),(-2/t,1)$, with $t > 0$; the lifted fourth gap is also one. Formula (6) gives $2/t, t, 2/t, t$. The opposite determinant is t, so $B = 2/t$ gives $-\langle t_u, t_v\rangle = 2t^2 = 8/B^2$. □

4 Exact defects and actual opposite-facet witnesses

We give the shared counting and geometric argument once, for a facet A^d of an ideal P^{d+1} with $d = 3$ or 4. Let $m = \mathrm{V}(A)$, N_i count outside vertices with i neighbors in A, and q count edges of A whose outgoing two-face is not a triangle.

Lemma 4.1 (Exact outside partition)**.** *With the notation above,*

$$N_1 = 2q, \qquad \mathrm{V}(P) = (1 + 2^{d-2})m + N_0 + q. \tag{8}$$

Proof. Capacity gives $i \le 2$ and $N_1 + 2N_2 = 2^{d-1}m$. There are $2^{d-2}m$ edges of A. For an edge uv the outgoing two-face is triangular exactly when $\langle h_u, h_v\rangle = -1$. Indeed $h_u + h_v$ is then a feasible future null vector. The three side halfplanes cut out the ideal triangle, and every other chamber halfspace contains its three vertices and their convex hull.

Conversely an outside vertex with two A-neighbors puts those neighbors at the two contact corners of one edge of its box, by the proof of Lemma 2.1. Its $d-1$ common vertical walls cut out the outgoing triangle, and A cuts its remaining side. The edge of A determines this face and its third vertex uniquely. Thus $N_2 = 2^{d-2}m - q$ is an equality, proving both formulas. □

Call a vertex of A *good* when all its $d-1$ opposite-pair coefficients relative to $t_u = n + h_u$ are < 2, and let G count them.

Lemma 4.2 (Good opposite facets)**.** *A good h_u is orthogonal to no other cusp-opposite facet h_v. Every vertex of h_u other than u and its 2^{d-1} neighbors belongs to N_0. At an N_0 vertex, at most two good h_u occur. Consequently*

$$G \le 2N_0 \quad (d=3), \qquad 9G \le 2N_0 \quad (d=4). \tag{9}$$

The witness conclusion remains true for a nongood h_u if its nonorthogonality to every other h_v has been proved separately.

Proof. If $h_u \perp h_v$, then $\langle t_u, t_v \rangle = -1$, so u, v are not adjacent by (5). They share at most $d-2$ facets of A. Otherwise among $d-1$ shared facets either an opposite pair is asymptotic only at u, or all are mutually orthogonal and their actual common one-face contains both u, v, making them adjacent. At least one opposite pair through u therefore misses v entirely. Its two products with t_v are strictly negative and at most -1 by the actual-wall gap. Pairing its sum αt_u with t_v forces $\alpha \ge 2$, a contradiction.

Let $w \in h_u$ be a nonneighbor of u. It lies outside A because the two supporting hyperplanes meet only at u. If w had an A-neighbor v, the outgoing edge wv would lie in h_v. At w the incident h_u, h_v cannot be orthogonal; they must be opposite walls in its rectangular box. With w at infinity, A's hemisphere is tangent to those parallel walls at the known actual points u, v (Figure 1). Its radius is half their separation and its tangency points have the same other coordinates. Since v is a box corner, u is also one. The geodesic uw is feasible against every wall by convexity; h_u and the other $d-1$ common vertical walls define it as an actual edge. This contradicts the choice of w.

For $d=3$, h_u has at least six vertices and degree four at u, leaving at least one witness. For $d=4$, Lemma 2.1 applied inside h_u gives at least $3 \cdot 6 = 18$ vertices; its eight neighbors at u leave at least nine witnesses. Any three distinct faces of a rectangular box include an orthogonal pair, so at most two good h_u can meet an N_0 vertex. Count these incidences. Only the first paragraph used goodness; the stated extension follows. □

5 A positive four-dimensional cusp defect

Theorem 5.1. *If R is a three-facet of an ideal right-angled four-polytope Q, then* $\mathrm{V}(Q) \ge 3\,\mathrm{V}(R) + 3$.

Proof. Put $m = \mathrm{V}(R)$. At a cusp of R the two pair coefficients α, β each occur twice among its four polygon corners. Every polygon has two small corners by Lemma 3.1. At least $\max(0, 8-2q)$ triangular faces have no exceptional edge, since an edge lies in two polygons; these have three small corners. Thus

$$S \ge 2(m+2) + \max(0, 8-2q), \qquad S \le 2m + 2G,$$

and $G \ge 2 + \max(0, 4-q)$. By Lemma 4.2, $2N_0 \ge G$. The exact defect is $\mathrm{V}(Q) - 3m = N_0 + q$, so it is at least

$$q + \left\lceil \frac{2 + \max(0, 4-q)}{2} \right\rceil \ge 3.$$

For $q = 0, 1, 2, 3$ the lower bounds are $3, 4, 4, 5$; for $q \ge 4$ the lower bound is at least $q + 1 \ge 5$. □

6 The eleven-vertex equality cannot occur

Lemma 6.1 (Six vertices force square cusps)**.** *An ideal right-angled three-polyhedron with six vertices has octahedral graph and square cusp rectangles.*

Proof. Its degree-four simple graph is the complete six-vertex graph minus a perfect matching. Euler gives eight triangular faces. Send a vertex u to infinity and its nonneighbor w to zero in upper half-space. The vertical walls at u bound $[x_-, x_+] \times [y_-, y_+]$, with zero strictly inside, and its four neighbors are the corners. The remaining facet across $x = x_+$ contains w and that side's two corners. Right angles put the hemisphere center on $x = x_+$; its other center coordinate is $(y_- + y_+)/2$ and its radius $(y_+ - y_-)/2$. Passing through zero gives $x_+^2 = -y_- y_+$. The opposite side gives the same equation for x_-^2, so $x_- = -x_+$. The horizontal pair similarly gives $y_- = -y_+$ and $y_+ = x_+$. The rectangle is square. This is a direct local proof, not a classification or rigidity assumption about Q. □

Theorem 6.2. *If an ideal right-angled four-polytope has a three-facet with eleven vertices, then it has at least thirty-seven vertices.*

Proof. Suppose equality in Theorem 5.1 gives $\mathrm{V}(Q) = 36$. Its integer defect inequalities force

$$q = 0, \qquad N_0 = 3, \qquad G = 6.$$

Every triangle of R has coefficients one. If T is their number, the small-corner bounds now give $2(11 + 2) + T \le S \le 2 \cdot 11 + 2 \cdot 6 = 34$. Equation (2) implies exactly eight triangles and five quadrilaterals, with no larger face. Equality also makes every nongood cusp have exactly one small coefficient and one coefficient at least two.

Each of the six good opposite facets h_u supplies at least one N_0 witness by Lemma 4.2, and at most two occur at any N_0 vertex. The total good incidence is exactly six. All vertices of a good h_u beyond u and its four neighbors are witnesses, so it has exactly six vertices. By Lemma 6.1, its cusp at u is square. Its four incident normals are precisely the same four normals used by R at u, at the same fixed t_u scale. Therefore

$$\alpha_u = \beta_u \quad \text{at each good vertex of } R. \tag{10}$$

No such six-vertex assertion has been made for a nongood facet.

We claim all eleven h_u are pairwise nonorthogonal. The good case is Lemma 4.2. Otherwise suppose $h_u \perp h_v$ with u nongood, and write $A < 2 \le B$ for its coefficients. Then $-\langle t_u, t_v \rangle = 1$, so u, v are nonadjacent. The two facet normals summing to At_u cannot both miss v, or their products with t_v would force $A \ge 2$. Thus v is on one of those polygonal facets through u. Its intrinsic corner coefficient at u is the OTHER coefficient B. It cannot be a triangle; it is a quadrilateral, and v, nonadjacent to u, is its opposite vertex. Lemma 3.2 forces $B = 2\sqrt{2}$, with coefficient $1/\sqrt{2}$ at its other two vertices.

At either of these two vertices, goodness gives both coefficients $1/\sqrt{2}$ by (10); nongoodness makes the other coefficient at least two. Neither coefficient is one. No triangle can meet it, so each vertex belongs to four quadrilateral facets. Only five quadrilateral facets exist. Their two four-element incident sets therefore share at least three. But among any three facets at a rectangular ideal cusp there is an opposite pair, whose hyperplanes share only that cusp. Two distinct vertices cannot share those three facets. This proves the claim.

Now apply the extended witness conclusion of Lemma 4.2 to ALL eleven h_u: each has a vertex beyond u and its four neighbors, which must lie in N_0 because its h_u is orthogonal to no other opposite facet. At an N_0 cusp at most two pairwise nonorthogonal facets occur. Thus $11 \le 2N_0 = 6$, the contradiction. See Figure 3. □

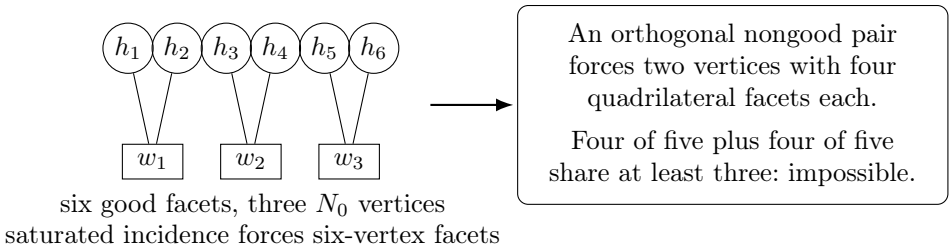


Figure 3: The equality argument counts only the six good facets before using their square cusps. The final witness count is extended to all eleven facets only after pairwise nonorthogonality is established.

7 Five-dimensional propagation

Theorem 7.1. *For every four-facet Q of an ideal right-angled five-polytope P, with $m = \mathrm{V}(Q)$ and $f = \mathrm{F}(Q)$,*

$$3\,\mathrm{V}(P) \geq 13m + 6f. \tag{11}$$

Proof. The exact outside partition is $\mathrm{V}(P) = 5m + N_0 + q$. The six intrinsic normals through a Q-cusp form three opposite pairs. Each pair coefficient occurs four times among its twelve polygon corners. Thus $S \leq 8m + 4G$.

The degree-eight graph and Euler give $r = 3m + f$ polygonal two-faces. Their corner incidence is $12m$, so their total defect beyond triangles is $12m - 3r = 3(m - f)$. Hence their triangle count satisfies $T \geq r - 3(m - f) = 4f$. An exceptional edge belongs to exactly three two-faces, so at least $\max(0, T - 3q)$ triangles are unexceptional. Lemma 3.1 consequently gives

$$S \geq 2r + \max(0, T - 3q) \geq 6m + 6f - 3q, \qquad 4G + 3q \geq 6f - 2m.$$

The $d = 4$ case of Lemma 4.2 gives $9G \leq 2N_0$. In particular $4G \leq 8N_0/9 \leq 3N_0$. Substitution into the last inequality yields $3(N_0 + q) \geq 6f - 2m$, proving $3\,\mathrm{V}(P) \geq 15m + 6f - 2m$. □

8 The carrier bridge and the dimension-five bounds

For an ideal right-angled four-polytope define

$$\kappa(Q) = 3\,\mathrm{V}(Q) - 4\,\mathrm{F}(Q) + 24.$$

We distinguish one internal companion theorem from published inputs.

Proposition 8.1 (Internal small-facet bridge)**.** *An entirely ideal right-angled four-polytope all of whose three-facets have at most ten vertices is an opposite-restricted chain of regular ideal* 24*-cells. A chain of k cells has $\kappa = 2 - 2k \leq 0$. Thus $\kappa(Q) > 0$ forces a three-facet with at least eleven vertices.*

This is the internal companion result, not a published classification of arbitrary four-polytopes. Its complete proof is in the unified ideal-four manuscript [4] and its named ten-vertex extension. The external small-dimensional input is the complete ideal right-angled three-polyhedron list through ten vertices: the octahedron, its triangular-face double, the square and pentagonal antiprisms, and the ten-vertex octahedrite in Kolpakov's Figure 7 [1], with the standard three-dimensional realization uniqueness. The internal proof first derives the antiprism and octahedrite cusp rectangles,

excludes them from this restricted facet spectrum by compatible box ratios and same-ridge neighbor constraints, and then constructs the octahedral sections and cuts proving the chain theorem. No unrestricted garland assertion is used. We cite that already written structural proof rather than duplicate it in a size-bound paper.

Lemma 8.2 (Positive carrier)**.** *Every ideal right-angled five-polytope has a four-facet Q with $\kappa(Q) > 0$. Such a facet satisfies $\mathrm{V}(Q) \geq 37$ and $\mathrm{F}(Q) \geq 25$.*

Proof. Write (v, e, r, g, f) for the five-polytope's face counts. Its four-box links give $e = 8v$, total three-face vertex incidence $24v$, and total four-facet vertex incidence $8v$. Every ideal right-angled three-face has two more facets than vertices; every two-face has three three- cofaces. Thus $3r = 24v + 2g$. Euler on the boundary four-sphere gives $g = 3v + 3f - 6$. Each three-face has two four-cofaces, and therefore

$$\sum_{Q} \kappa(Q) = 3(8v) - 4(2g) + 24f = 48. \tag{12}$$

Choose a positive term and apply Proposition 8.1. It contains a three-facet R with $M \geq 11$ vertices. If $M = 11$, Theorem 6.2 gives $\mathrm{V}(Q) \geq 37$; if $M \geq 12$, Theorem 5.1 already gives $\mathrm{V}(Q) \geq 39$. Separately, Lemma 1.2 applied to R gives

$$\mathrm{F}(Q) \geq 1 + \mathrm{F}(R) + \mathrm{V}(R) = 2M + 3. \tag{13}$$

□

The separate facet count is not inferred from a vertex count.

Theorem 8.3. *Every entirely ideal finite-volume right-angled five-polytope satisfies*

$$\mathrm{F}(P) \geq 63, \qquad \mathrm{V}(P) \geq 211.$$

Every entirely ideal right-angled six-polytope has at least 275 facets.

Proof. At the positive carrier, Lemma 1.2 gives $\mathrm{F}(P) \geq 1 + 25 + 37 = 63$. Theorem 7.1 gives $3\,\mathrm{V}(P) \geq 13 \cdot 37 + 6 \cdot 25 = 631$, hence $\mathrm{V}(P) \geq 211$. Every five-facet of an ideal six-polytope satisfies both inequalities; another facet injection gives $\mathrm{F}(P^6) \geq 1 + 63 + 211 = 275$. □

9 Good-edge transport in dimension six

We next prove a universal relation between the six-dimensional face counts. No arithmetic hypothesis or facet spectrum enters this section. At an ideal cusp choose future null vectors v, z with $\langle v, z \rangle = -1$ and orthonormal positive $e_1, \ldots, e_5$ perpendicular to them. Centering its box gives the ten outward unit normals

$$n_i^{\pm} = \pm e_i + \frac{w_i}{2} v, \qquad w_i > 0. \tag{14}$$

A sign vector $\epsilon \in \{\pm 1\}^5$ specifies an actual edge through the cusp. Call it good if all five two-faces containing it are triangles, and call the cusp cubic if all its w_i agree.

Put $S = \sum_i w_i^2$. The third wall of the triangle obtained by dropping the ith edge-support wall is forced to be

$$c_i^{\epsilon} = \frac{S - 2w_i^2}{4w_i} v + \frac{2}{w_i} z + \sum_{j \neq i} \epsilon_j \frac{w_j}{w_i} e_j. \tag{15}$$

Indeed its products with both n_i^+, n_i^- are -1, its products with the other four supporting normals are zero, and its square is one. These are actual ambient facet normals: right angles make the wall cutting the polygon side orthogonal to its supporting walls.

At a good edge the five c_i^ϵ are mutually orthogonal and

$$n_i^{\epsilon_i} + c_i^\epsilon = \frac{v_\epsilon}{w_i}, \qquad v_\epsilon = \frac{S}{4}v + 2z + \sum_j \epsilon_j w_j e_j. \tag{16}$$

Direct products give $v_\epsilon^2 = 0$ and $\langle v, v_\epsilon \rangle = -2$. Its positive-sum expression makes it feasible against every other actual wall; against its summands its product is zero. It lies on the same supporting edge as v and is its second ideal endpoint. The ten distinct displayed normals fill its box link. Their opposite sums in this normalization are v_ϵ/w_i, so its new widths are $1/w_i$. Good edges preserve cubicity in both directions.

Lemma 9.1. *At a noncubic cusp at most sixteen of its thirty-two edges are good. If g counts good edges and t triangular two-faces at a cusp, then $t \le 64 + g/2$, and therefore $t \le 72$ at a noncubic cusp.*

Proof. Suppose two good sign vectors differ only in direction k. For $i \neq k$ their two third walls have product $1 - 2w_k^2/w_i^2$. Both are orthogonal to the common c_k and the three unchanged supporting normals. Product zero would give six mutually orthogonal actual walls and a finite vertex. The gap of Lemma 1.1 forces $1 - 2w_k^2/w_i^2 \le -1$, so $w_k \ge w_i$ for every i. At a noncubic cusp a nonmaximal direction pairs all thirty-two signs and at most one member of each pair is good. Thus $g \le 16$. The five-cube has eighty edges. Every nongood cube vertex touches a nontriangular link edge, so $2(80 - t) \ge 32 - g$. □

At a cubic cusp normalize $w_i = 1$. Then v is canonical, the sum of any opposite pair of actual unit normals. The vector

$$h = z - v/8, \qquad h^2 = 1/4, \qquad \langle h, v \rangle = -1$$

is independent of signed permutations of the centered frame. Across a good edge define $e_i' = n_i^{\epsilon_i} - v_\epsilon/2$ and $z' = h + v_\epsilon/8$. Their products give a new centered orthonormal frame, $z'^2 = 0$, $\langle z', v_\epsilon \rangle = -1$, and $h = z' - v_\epsilon/8$. Thus h is constant on each finite component of cubic cusps connected by good edges, including around cycles.

Choose a future null $Z \perp h$ in each component. Set $E(v) = -\langle Z, v \rangle > 0$ and $p_i = \langle Z, e_i \rangle / E(v)$. The energy is nonzero because $Z \perp h$ but $\langle v, h \rangle = -1$. The equations $Z^2 = \langle Z, h \rangle = 0$ give

$$\sum_i p_i^2 = \frac{1}{4}, \qquad \frac{E(v_\epsilon)}{E(v)} = \frac{3}{2} - \sum_i \epsilon_i p_i. \tag{17}$$

Every formal ratio is positive even when that sign vector is not an actual good edge: its formal endpoint is future null and has h-product -1, so it is not parallel to Z.

10 Sharp reciprocal averaging and six-dimensional vertices

Lemma 10.1. *For independent uniform signs and $\sum_{i=1}^5 p_i^2 = 1/4$,*

$$\mathbb{E} \frac{1}{3/2 - \sum_i \epsilon_i p_i} \le \frac{103}{132}. \tag{18}$$

Equality holds exactly when all $p_i^2 = 1/20$.

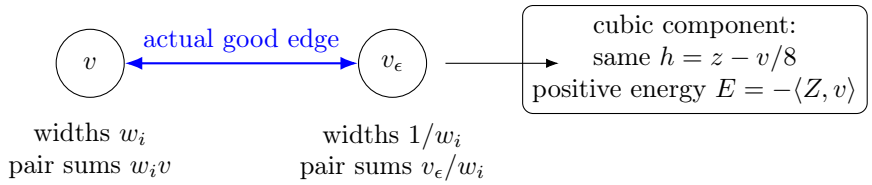


Figure 4: Actual endpoint transport and the unweighted reciprocal-pair sum. Width transport is exact; the thirty-two formal directions are used only as an upper relaxation after every ratio is positive.

Proof. Put $a_i = p_i^2$, $X = \sum_i \sqrt{a_i}\epsilon_i$, $\mu = 3/2$. On the compact simplex $a_i \geq 0$, $\sum_i a_i = 1/4$, one has $|X| \leq \sqrt{5}/2 < \mu$. Sign symmetry and a uniformly convergent positive series give

$$\mathbb{E}(\mu - X)^{-1} = \mu^{-1} \sum_{k \geq 0} \mu^{-2k} \mathbb{E} X^{2k}.$$

Fix two squares a, b, their sum s, and the remaining independent symmetric sum R. For $Y = \sqrt{a}\epsilon + \sqrt{b}\eta$,

$$\mathbb{E} Y^{2j} = \sum_{\ell=0}^{\lfloor j/2 \rfloor} \binom{j}{2\ell} s^{j-2\ell} (4ab)^{\ell}.$$

Balancing a, b increases every even moment. Expanding $\mathbb{E}(R + Y)^{2k}$ has only nonnegative even terms, so it too increases. The fourth moment increases strictly by $(a - b)^2$ if $a \neq b$. The reciprocal average is continuous and has a compact maximizer, including on the boundary; no maximizer can have unequal squares. At equality, $X^2 = 5/4, 9/20, 1/20$ with multiplicities $2, 10, 20$. Pairing opposite signs gives exactly

$$\frac{3}{2}\left(\frac{1/16}{1} + \frac{5/16}{9/5} + \frac{5/8}{11/5}\right) = \frac{103}{132}.$$

This proves the global inequality and its equality characterization, not merely a sampled finite-direction maximum. □

At a cubic vertex the positive function $1/E$ has outgoing ratio sum at most $32 \cdot 103/132 = 824/33$, allowing all formal neighbors. Summing over each good-edge component, every actual undirected edge contributes a reciprocal pair at least two. The unweighted total good degree of cubic cusps is therefore at most $(824/33)V_c$. Lemma 9.1 then bounds their triangular corners by $(64 + 412/33)V_c = (2524/33)V_c$; the noncubic bound 72 is smaller. If T is the number of triangular two-faces of P^6, each has three cusps, so

$$3T \leq \frac{2524}{33} V, \qquad V = \mathrm{V}(P^6). \tag{19}$$

Theorem 10.2. *Every entirely ideal finite-volume right-angled six-polytope with F facets and V vertices satisfies*

$$F \geq 275, \qquad V \geq \left\lceil \frac{6039F}{272} \right\rceil \geq 6106.$$

Proof. Write $a_j = f_j(P)$, $a_0 = V$, $a_5 = F$. Its box links give $a_1 = 16V$. Summing the five-dimensional identity $g = 3v + 3f - 6$ over facets gives $3a_3 = 30V + 6a_4 - 6F$; Euler gives $a_2 = 25V + a_4 - F$. Polygon-cusp incidence is $80V$. Since nontriangles have at least four corners,

$$T \geq 4a_2 - 80V = 20V + 4(a_4 - F).$$

Together with (19), multiplying this lower bound by THREE gives $a_4 - F \leq 136V/99$. Each five-facet has at least sixty-three four-facets by Theorem 8.3; every four-face has two five-cofaces. Thus

$$\frac{61}{2}F \leq a_4 - F \leq \frac{136}{99}V, \qquad 272V \geq 6039F.$$

The facet bound $F \geq 275$ was already proved. Finally $6039 \cdot 275 = 1660725$, whereas $272 \cdot 6105 = 1660560 < 1660725 \leq 1660832 = 272 \cdot 6106$. □

The reciprocal constant is sharp for formal cusp-frame averages. No attainment by an actual polytope, or optimality of the size bounds, is claimed.

11 A restriction on mixed-dimensional boundaries

Here the additional premises are published. The strict Nikulin–Khovanskii inequality for polytopes simple at edges is

$$\frac{1}{f_k(P)} \sum_{E \in \Omega_k(P)} f_l(E) < \binom{n-l}{n-k} \frac{\binom{\lfloor n/2 \rfloor}{l} + \binom{\lceil n/2 \rceil}{l}}{\binom{\lfloor n/2 \rfloor}{k} + \binom{\lceil n/2 \rceil}{k}}, \quad 0 \leq l < k \leq \lceil n/2 \rceil.$$

The right-angled hypotheses and this precise range are recorded in Alexandrov [2, §§2,6] and Egorov [3]. For $n = 11, 12$, the permitted choices $(k, l) = (5, 4), (6, 5)$ give strict averages twenty and forty-two. Faces of dimension at least three remain finite-volume and right-angled, so both applications are valid even with mixed vertices.

The general five-dimensional facet minimum sixteen is Dufour's $f_4 \geq 1 + \nu_4$ with $\nu_4 = 15$, stated in [3, Lemma 2.3]. The six-dimensional minimum twenty-seven is Dufour's $f_5 \geq 1 + \nu_5$ with $\nu_5 \geq 26$, stated in [2, Proposition 6.5]. Their noncompact scope covers these dimensions because compact right-angled polytopes have dimension at most four, by the published Potyagailo–Vinberg theorem. The original subscription Dufour proof was not independently available here; we use these precisely stated premises, not an asserted new proof of that theorem.

Let N_k count all k-faces and I_k the entirely ideal ones. Our new necessary counts give

$$16N_5 + 47I_5 \leq \sum_{E \in \Omega_5(P)} f_4(E) < 20N_5, \qquad \frac{I_5}{N_5} < \frac{4}{47},$$

and

$$27N_6 + 248I_6 \leq \sum_{E \in \Omega_6(P)} f_5(E) < 42N_6, \qquad \frac{I_6}{N_6} < \frac{15}{248}.$$

They are coupled necessary incidence restrictions, not contradictions. In particular the right-angled dimension ceilings remain six for the entirely ideal class and twelve for the general finite-volume class.

12 Proof interfaces and verification scope

The small-facet bridge of Proposition 8.1 is an internal companion theorem with explicit published three-dimensional inputs; it is not an external axiom or a classification of arbitrary ideal four-polytopes. The outside capacity, fixed-scale spinor proof, positive defect, eleven-vertex equality exclusion, propagation, actual good-edge transport, and all-moment reciprocal proof are human proofs given in this paper. Their separate internal mathematical reviews are indexed in the preparation map accompanying this source.

Existing exact scripts check frame identities, grouped sign sums and older rational consequences. Such finite checks do not replace the universal proofs for every width or polygon. Existing conditional Lean modules cover earlier arithmetic interfaces with their geometric premises explicit. No full formal proof of the new geometric chain or of the 63/211 and 275/6106 constants is claimed by this revision. Source reviews, exact execution records and typesetting records are kept separately; successful typesetting is not a mathematical proof. See `docs/ideal_right_angled_six_face_constraints_preparation_20261019.md` for the source archive, theorem map, frozen dependencies and execution scope.

References

[1] A. Kolpakov, *On the optimality of the ideal right-angled 24-cell*, Algebraic & Geometric Topology 12 (2012), 1941–1960. `https://doi.org/10.2140/agt.2012.12.1941`.

[2] S. Alexandrov, *On ideal vertices of right-angled hyperbolic polyhedra*, 2023. `https://arxiv.org/abs/2303.09533`.

[3] A. Egorov, *On lower bounds for the number of ideal and finite vertices of right-angled hyperbolic polyhedra in dimensions from 5 to 12*, 2026. `https://arxiv.org/abs/2603.29925`.

[4] Coxeter Polytopes Frontier Project, *Volume and structure of ideal right-angled four-polytopes*, internal companion manuscript, `manuscripts/ideal_ra4_volume_and_structure.tex`, with the separately reviewed extension `results/ideal_ra4_ten_vertex_facet_exclusion_research_20261019.md`. The three-dimensional small-facet classification is the stated external input; the four-dimensional structural conclusions are internal proofs.